ENGINEERING PORTFOLIO

Rishabh Bavithran

Mechatronics & Controls Engineer · Systems Integration · Automation

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OVERVIEW

Skills & Tools

Languages & Firmware

- C++, Python, C, MATLAB (Simulink, Simscape)
- Microcontrollers: NXP, Arduino, ESP32
- SBCs: Raspberry Pi, Jetson Nano / Orin Nano

Comms & Controls

- Modbus (RTU/TCP), I2C, SPI, UART
- RS-232/RS-485 serial, DAQ systems
- ROS2, P&ID design, control logic

Fluid & Process Systems

- Pneumatic actuators, mass flow controllers
- Fluid control systems, gas delivery design
- Sterilization & materials compatibility

Fabrication & CAD

- Fusion 360, SolidWorks, CREO Parametric
- Design for CNC manufacturing (vendor-fabricated), laser cutting, 3D printing
- Soldering, wiring, electromechanical build

Infrastructure & Ops

- Git, Tailscale, SSH - remote deployment
- Lucidchart / P&ID documentation
- Linux (Ubuntu), embedded Linux on ARM64

Domain Depth

- Bioreactor / gas-fermentation systems
- Sensor interlocks & safety systems
- Multi-agent robotics, MPC, path planning

PROFESSIONAL WORK

CarbonBridge - DGF-100 Automated Bioreactor

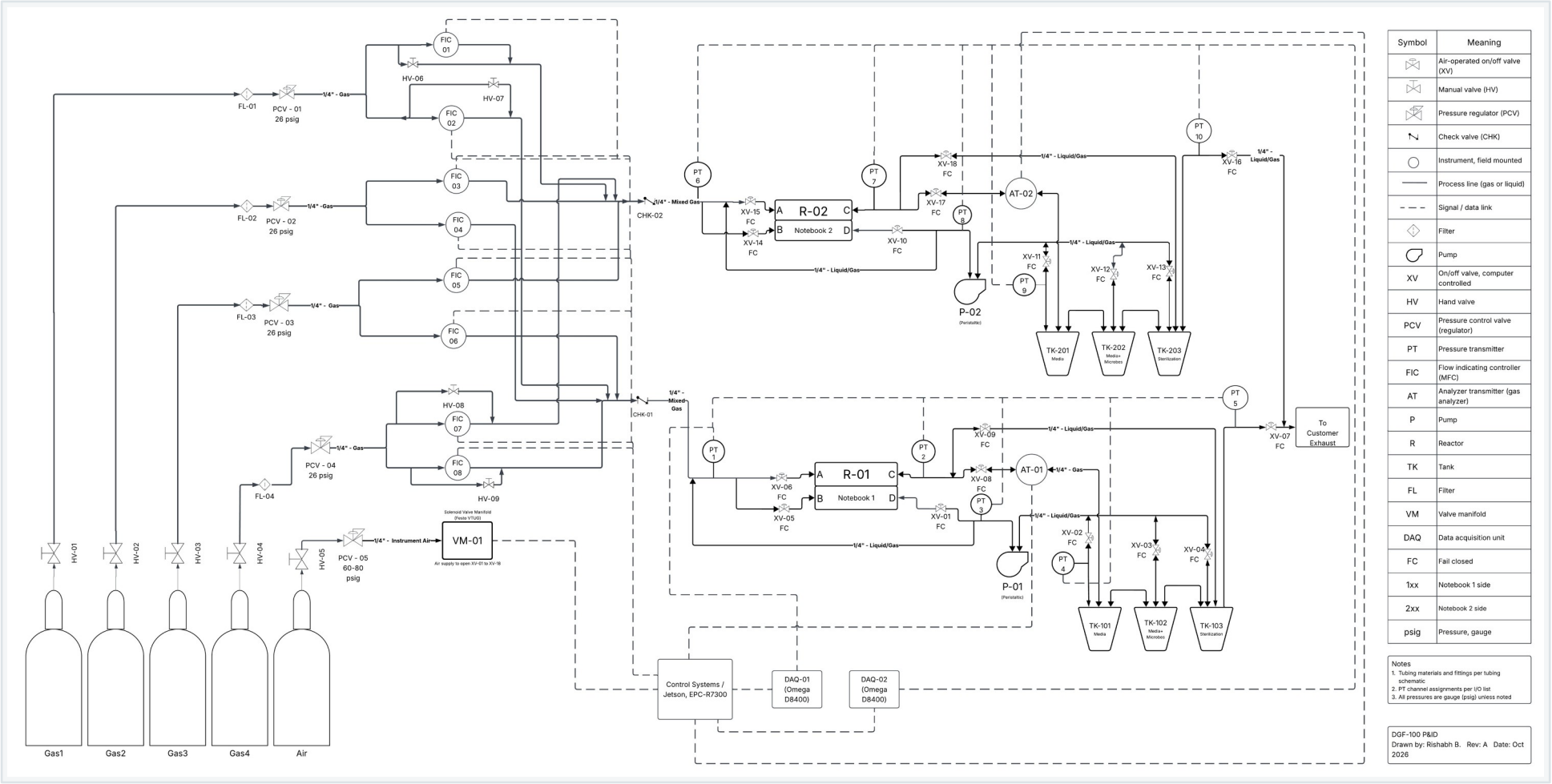
Lead Engineer (sole on-site) · Nov 2024 – Present

- Independently conceived, designed, and built the DGF-100 automated gas-fermentation reactor system - sold to Lawrence Berkeley National Lab and A*STAR Singapore, in daily use by the internal science team.
- Designed the core reactor vessel from first principles in SolidWorks/Fusion 360
- Authored the governing tubing schematic and control logic - microbe delivery, fluid routing, valve actuation - before any wiring or code existed; later formalized it into an ISA-5.1 P&ID. Sequenced 18 solenoid valves across 50+ named states.
- De-risked manually first, achieving a 16x methanol yield increase (8mg → 128mg) on a 500cm² prototype before automating.
- Built the full electrical wiring schematic and BOM; performed the entire electromechanical build solo - tube bending, wiring, soldering, assembly.
- Authored the complete C++ automation stack from scratch: per-device HAL classes (Modbus TCP/RTU, RS-232), concurrent safety threads, JSON-driven config, a BIST/leak-test suite, and remote deployment via Git/Tailscale/SSH.



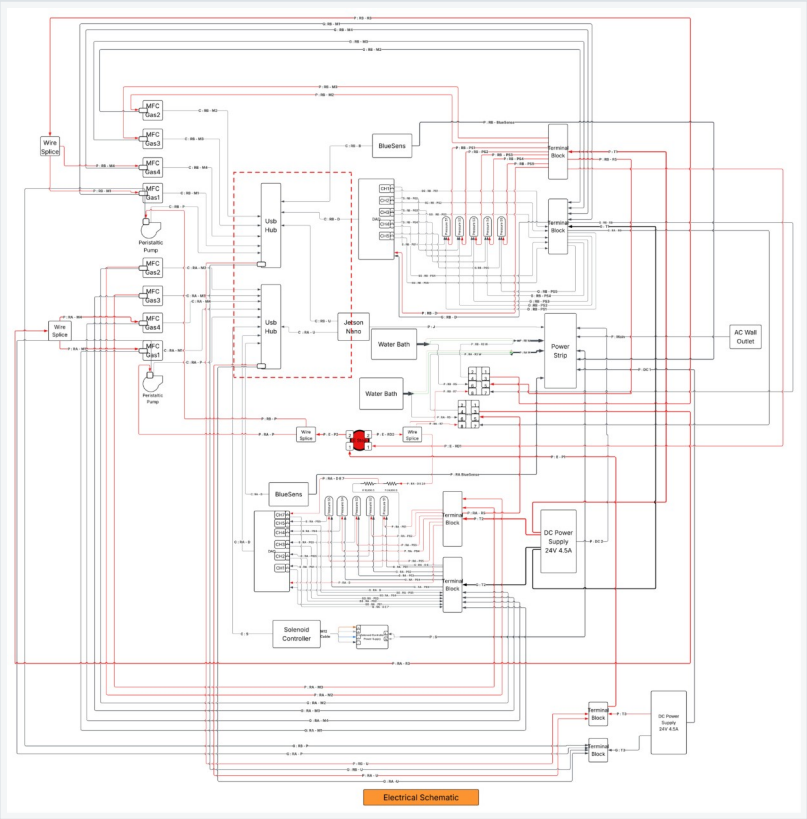
Full DGF-100 system - MFCs, peristaltic pumps, water baths, live control UI

Governing Process Diagram: P&ID

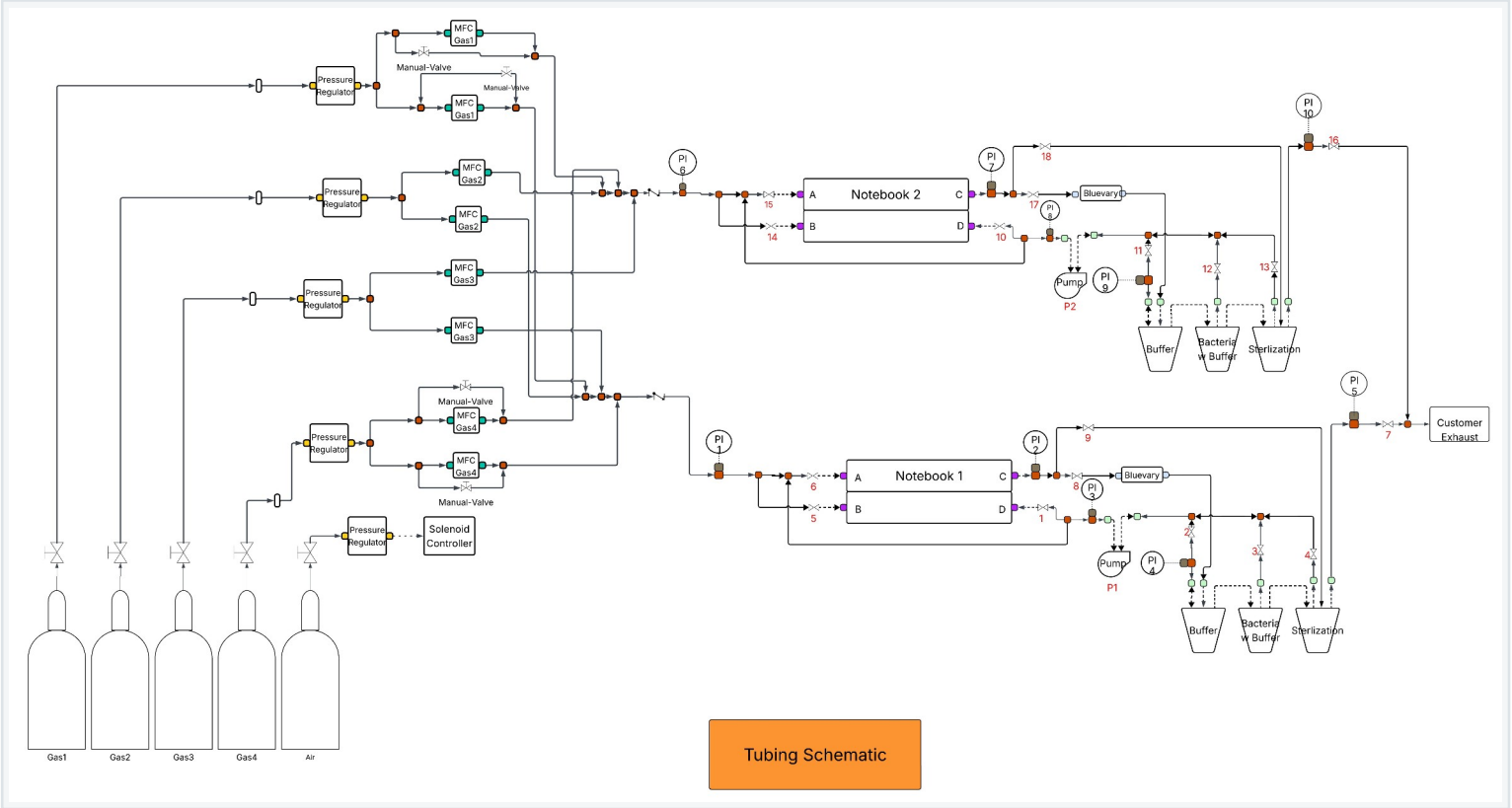


P&ID - dual reactor trains, gas supply, instrumentation, and control signals

Build Documents: Electrical & Tubing



Electrical & communications schematic - dual reactor trains, DAQ, solenoid control, E-stop

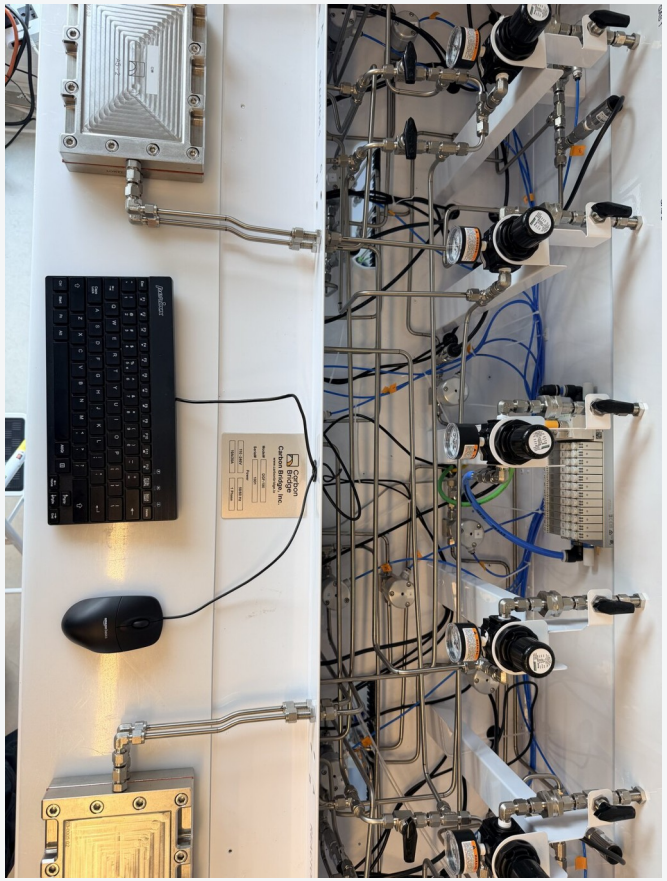


Tubing schematic - every fitting and part linked, used for assembly

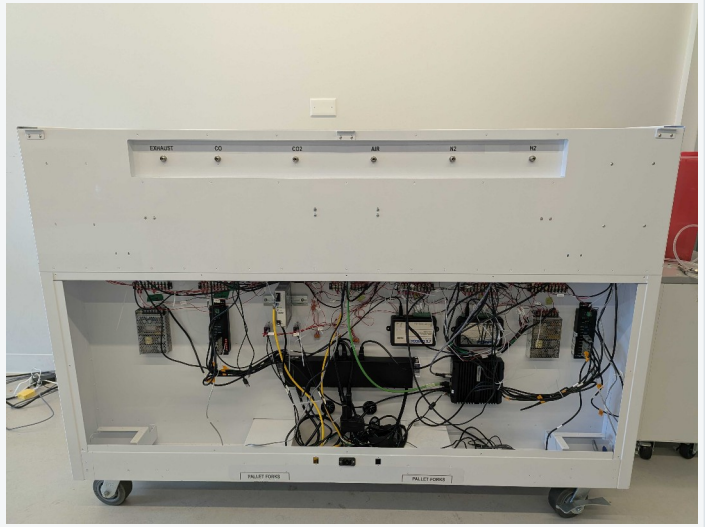
The Physical Build



Gas manifold, regulators & reactor vessel



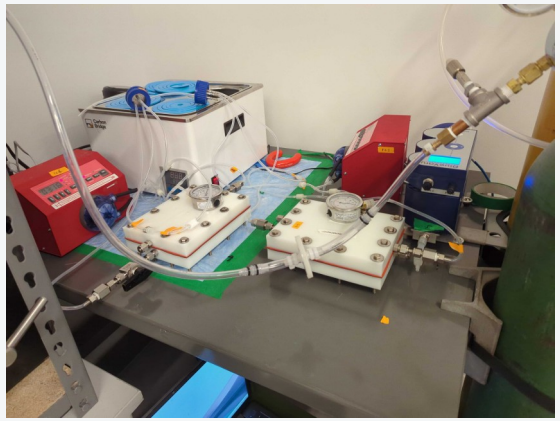
Dual reactor bay with nameplate & controls



Rear panel - labeled gas ports, wiring bay

Complete electromechanical build performed solo - tube bending, wiring, soldering, and final assembly of every subsystem shown here.

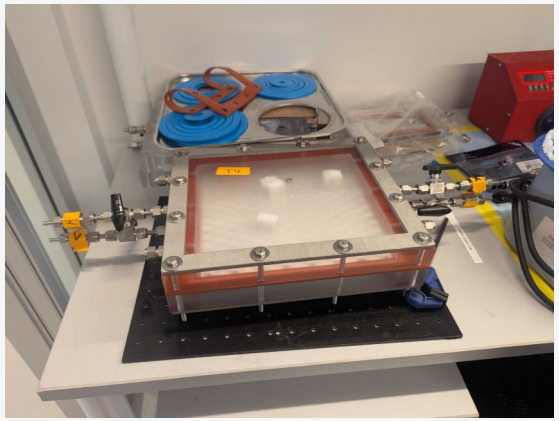
From Bench Prototype to Automated System



FEB – APR 2025

Bench-Scale Testing

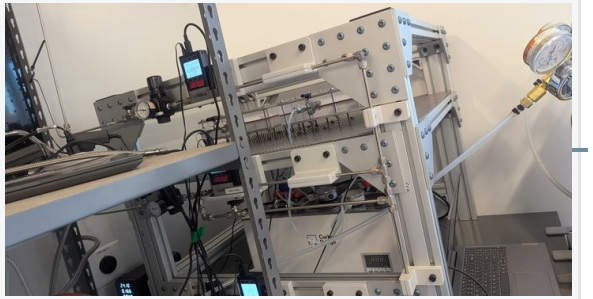
First scale-up from a 10cm² component; manual gas & leak checks on flat 'notebook' reactors.



APR – AUG 2025

Full-Size Prototype

4x-scale vessel proved out core fluid mechanics & sealing design - 16x methanol yield gain (8mg→128mg).



AUG – NOV 2025

Reactor Shelf

SolidWorks-designed aluminum-extrusion rack, custom 3D-printed mounts, for parallel manual testing.



NOV 2025 – PRESENT

DGF-100 (Automated)

Same reactor architecture carried forward into a fully automated, remotely deployed system.

Diagnostic Depth: Two Debugging Stories

Shared Solenoid Hardware - Race Condition Across Reactor Threads

- Context: both reactors (A and B) run as separate threads from the same shared reactor struct - each owns its own MFCs, pressure sensors, and valve addresses - except the solenoid controller, which is a single physical unit shared by both.
- Symptom: intermittent core dump when changing valve states, mid-experiment - the first bug ever caught on the DGF-100.
- Reasoning: since every other component was fully independent per-reactor, the fault had to be something the two threads actually shared, not a bug in either reactor's own logic.
- Investigation: traced it to both reactor threads issuing valve commands to the same physical solenoid hardware at the same time - a race condition, not a per-reactor fault.
- Fix: introduced a mutex to lock access to the shared solenoid hardware, serializing valve commands across both threads.

Valve Sequencing Failure (18-Valve Manifold)

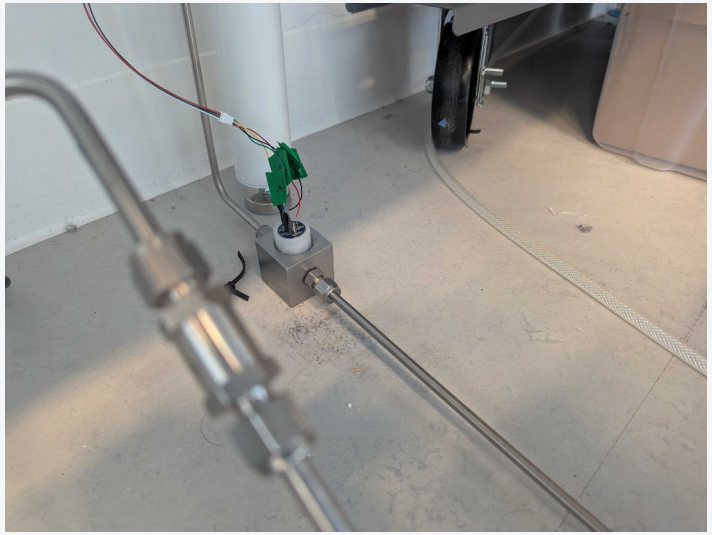
- Context: the DGF-100 sequences 18 solenoid valves across 50+ named states via Modbus commands to a Festo VTUG solenoid block, controlled from the C++ automation stack
- Symptom: each of the 18 solenoid valves actuated fine alone; wired into the control loop, the first state change worked, the second silently failed.
- Reasoning: since isolated actuation was correct, the fault was specific to back-to-back state changes.
- Investigation: reproduced directly in test code - two rapid state changes reliably failed, confirming a sequencing issue.
- Root cause: the Festo hardware auto-closed the Modbus connection after every command - a second write hit an already-closed connection.
- Fix: built a resilience wrapper that reopens the connection per command.

Instrumentation & Process Improvements



Gas Delivery Redesign

Rebuilt the microbe growth chambers' gas delivery system: modular pegboard-mounted layout with labeled Alicat MFCs (Methane, CO2/H2, Air/O2), custom 3D-printed brackets, and reorganized tubing runs.



Methane Sensor Integration

Designed a stainless steel housing for a UART-based methane sensor at the growth chamber outlet; wrote the Python interface and calibration routine.



Temperature Troubleshooting

When microbe growth stalled, instrumented each growth chamber with K-type thermocouples on an Omega DAQ to continuously log temperature (methanotrophs need ~30°C) - ruling temperature in or out as a root cause.

PROFESSIONAL WORK

Earlier Roles

Azul Bio

Aug 2024 – Oct 2024 · New York

- Coral microbiome booster systems - ESP32/PCB-based embedded hardware.
- Hands-on electronics work spanning sensor integration and board-level troubleshooting for an early-stage biotech hardware platform.

Void Robotics

Jun 2023 – Aug 2023 · Remote

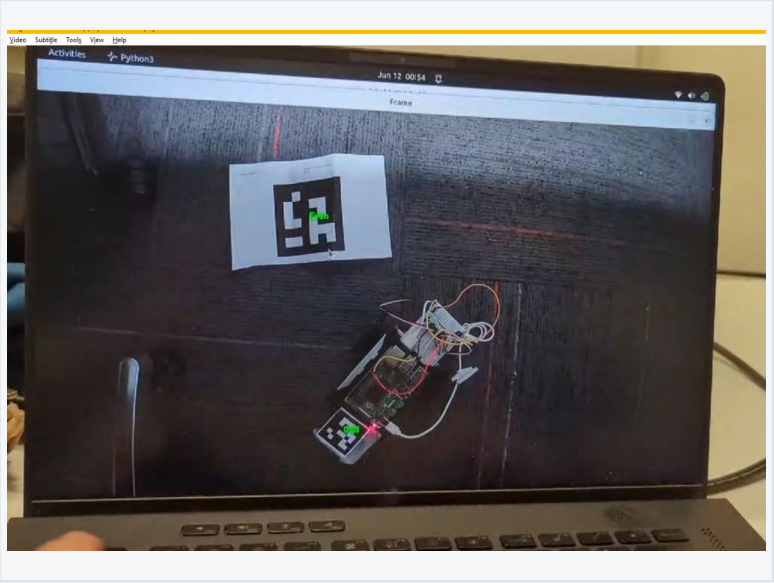
- ROS2/Blynk-based delivery robot.
- Contributed to the robot's control and connectivity stack, working across embedded firmware and app-level control (Blynk) for remote operation.

Both roles reinforced a pattern that carried into CarbonBridge: picking up new hardware/software ecosystems out of necessity.

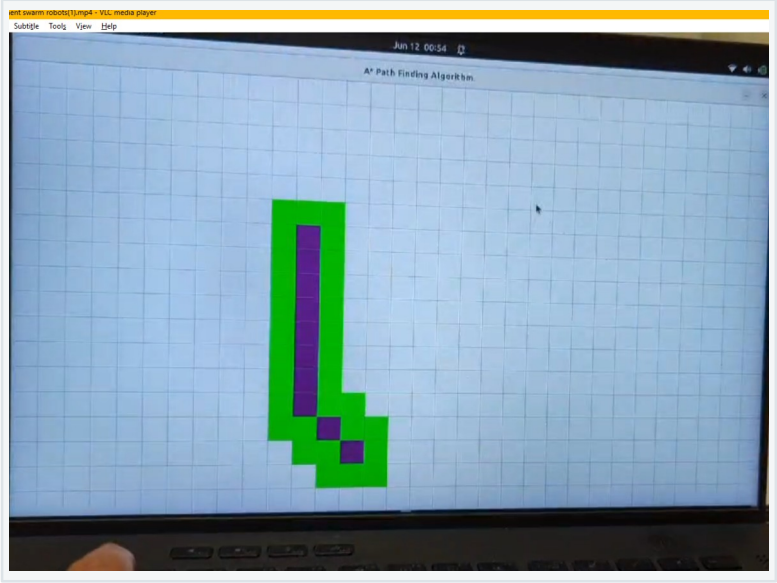
Also: Perfint Healthcare (Dec 2020 – Mar 2021) - end-effector design for a spine surgery robot.

MPC Swarm - Autonomous Warehouse Management

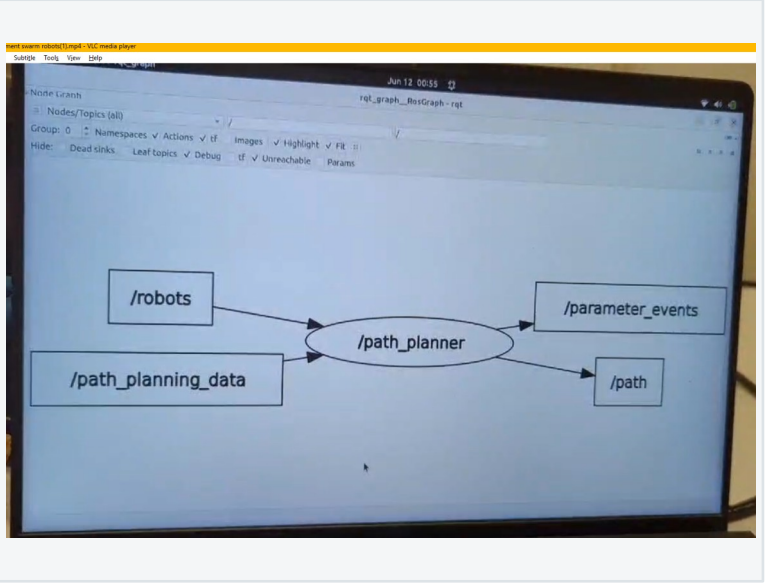
- Overhead-camera system tracking multiple robots via ArUco tags, each picking/placing at an assigned location.
- Time-augmented A* path planning - treating time as an extra dimension so the planner detects and resolves inter-robot path conflicts, not just spatial collisions.
- ROS2 architecture: each robot's node publishes position updates; a master node continuously checks for conflicts across the fleet and triggers replanning.



ArUco tag detection on a physical robot

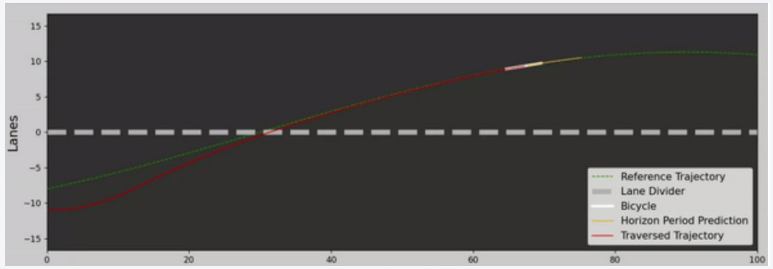


A planned path from live camera detection*

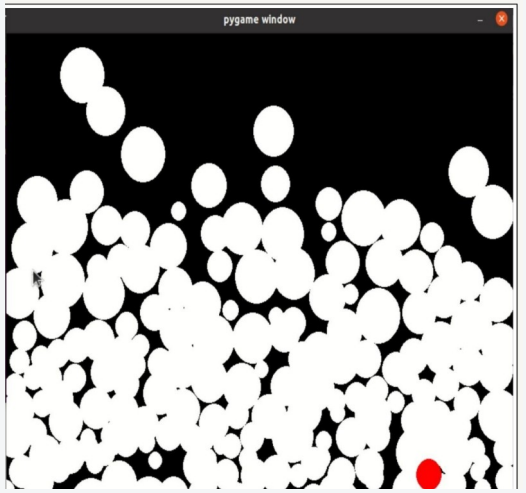


ROS2 node graph - planner architecture

Controls & Simulation Coursework



DH Table → Jacobian → Inverse Dynamics → PID



Bicycle Path Tracking (MPC)

- Modeled kinematic bicycle dynamics; implemented an MPC for reference-path tracking.
- Tracked a lane-change trajectory, visualizing the receding-horizon prediction against the traversed path.

Robotic Manipulator Path Planning

- Derived the manipulator's Jacobian and forward-kinematics relations from its DH table.
- Built a Simulink model of a second-order inverse-kinematic control law with PID feedback.
- Modeled a trapezoidal-velocity motion profile under inverse dynamic control.

Micro-Swarm Robots - Brazil Nut Effect

- Simulated 200 micro-swarm robots in ROS2, modeling particle dynamics for granular transport.
- Modeled emergent particle stratification; built a neighbor-data estimation method sizing foreign particles to 98% accuracy.

Completed jointly with a fellow NYU M.S. classmate as part of shared coursework.